

Name: AAVISHKAR SINGH AAVISHKAR SINGH | Reg. No: RA2511003011024

Unit 3 : Tutorial 2 — Signed, Unsigned and Fast Multiplication

■ Section A: Unsigned Multiplication

- 1a) 1101×1011 : partial products $1101 + 0000 + 110100 + 1101000 \rightarrow 10001111$. Answer: 10001111 ($13 \times 11 = 143$)
- 1b) 10101×0110 : partial products $000000 + 101010 + 1010100 + 0000000 \rightarrow 1111110$. Answer: 1111110 ($21 \times 6 = 126$)
- 2a) 12×5 : $12 = 1100$, $5 = 0101$; binary product $1100 \times 0101 = 111100$; back to decimal: $32 + 16 + 8 + 4 = 60$. Answer: 60
- 2b) 23×9 : $23 = 10111$, $9 = 1001$; binary product = 11001111 ; decimal: $128 + 64 + 8 + 4 + 2 + 1 = 207$. Answer: 207
- 3) 1110 (M) $\times$ 1001 (Q), 4-bit registers, add-and-shift hardware algorithm:
 - Init: $A = 0000$, $MQ = 1001$, $MD = 1110$, counter = 4
 - Step 0: $MQ_0 = 1 \rightarrow A = A + MD = 0000 + 1110 = 1110$ (product $1110\ 1001$); logical shift right $\rightarrow 0111\ 0100$
 - Step 1: $MQ_0 = 0 \rightarrow$ no add; shift $\rightarrow 0011\ 1010$
 - Step 2: $MQ_0 = 0 \rightarrow$ no add; shift $\rightarrow 0001\ 1101$
 - Step 3: $MQ_0 = 1 \rightarrow A = A + MD = 0001 + 1110 = 1111$ ($1111\ 1101$); final shift $\rightarrow 0111\ 1110$
 - Product = $01111110 = 126$ (check: $14 \times 9 = 126$ ✓)

■ Section B: Signed Multiplication (Booth's Algorithm)

- a) $(+7) \times (+3)$: $M = 0111$, $-M = 1001$, $Q = 0011$, $A = 0000$, $Q-1 = 0$
 - Cycle 1: $Q_0Q-1 = 10 \rightarrow A = A - M = 1001$; shift $\rightarrow A = 1100$, $Q = 1001$, $Q-1 = 1$
 - Cycle 2: $Q_0Q-1 = 11 \rightarrow$ no op; shift $\rightarrow A = 1110$, $Q = 0100$, $Q-1 = 1$
 - Cycle 3: $Q_0Q-1 = 01 \rightarrow A = A + M = 0101$; shift $\rightarrow A = 0010$, $Q = 1010$, $Q-1 = 0$
 - Cycle 4: $Q_0Q-1 = 00 \rightarrow$ no op; shift $\rightarrow A = 0001$, $Q = 0101$, $Q-1 = 0$
 - $A + Q = 0001\ 0101 = 00010101 = 21$ ($7 \times 3 = 21$ ✓)
- b) $(-5) \times (+4)$: $M = 1011$, $-M = 0101$, $Q = 0100$, $A = 0000$, $Q-1 = 0$
 - Cycle 1: $Q_0Q-1 = 00 \rightarrow$ shift $\rightarrow A = 0000$, $Q = 0010$, $Q-1 = 0$
 - Cycle 2: $Q_0Q-1 = 00 \rightarrow$ shift $\rightarrow A = 0000$, $Q = 0001$, $Q-1 = 0$
 - Cycle 3: $Q_0Q-1 = 10 \rightarrow A = A - M = 0000 + 0101 = 0101$; shift $\rightarrow A = 0010$, $Q = 1000$, $Q-1 = 1$
 - Cycle 4: $Q_0Q-1 = 01 \rightarrow A = A + M = 0010 + 1011 = 1101$; shift $\rightarrow A = 1110$, $Q = 1100$, $Q-1 = 0$
 - $A + Q = 1110\ 1100 = -20$ in 2's complement ($-5 \times 4 = -20$ ✓)
- c) $(-7) \times (-3)$ with 5-bit registers: $M = 10001$, $-M = 01111$, $Q = 11101$, $A = 00000$, $Q-1 = 0$
 - Cycle 1: $10 \rightarrow A = A - M = 01111$; shift $\rightarrow A = 00111$, $Q = 11110$, $Q-1 = 1$
 - Cycle 2: $01 \rightarrow A = A + M = 10110$; shift $\rightarrow A = 11011$, $Q = 01111$, $Q-1 = 0$
 - Cycle 3: $10 \rightarrow A = A - M = 01010$; shift $\rightarrow A = 00101$, $Q = 00111$, $Q-1 = 1$
 - Cycle 4: $11 \rightarrow$ no op; shift $\rightarrow A = 00010$, $Q = 10011$, $Q-1 = 1$
 - Cycle 5: $11 \rightarrow$ no op; shift $\rightarrow A = 00001$, $Q = 01001$, $Q-1 = 1$
 - Product (A, Q) = $00001\ 01001 = 000010101 = +21$ ($-7 \times -3 = +21$ ✓)
- d) $M = 110011$ (-13), $Q = 101110$ (-18): $A = 000000$, $-M = 001101$, $Q-1 = 0$

- Cycle 1: 00 → no op; shift → $A = 000000$, $Q = 010111$, $Q-1 = 0$
- Cycle 2: 10 → $A = A - M = 001101$; shift → $A = 000110$, $Q = 101011$, $Q-1 = 1$
- Cycle 3: 11 → no op; shift → $A = 000011$, $Q = 010101$, $Q-1 = 1$
- Cycle 4: 11 → no op; shift → $A = 000001$, $Q = 101010$, $Q-1 = 1$
- Cycle 5: 01 → $A = A + M = 110100$; shift → $A = 111010$, $Q = 010101$, $Q-1 = 0$
- Cycle 6: 10 → $A = A - M = 000111$; shift → $A = 000011$, $Q = 101010$, $Q-1 = 1$
- Product = $000011\ 101010 = 000011101010 = +234$ ($-13 \times -18 = +234$ ✓)

■ Section C: Fast Multiplication (Karatsuba)

- a) 1011×1101 (4-bit): split $X = 10|11$, $Y = 11|01$ (halves of 2 bits)
- $z_2 = X_h \cdot Y_h = 10 \times 11 = 110$ (6); $z_0 = X_l \cdot Y_l = 11 \times 01 = 11$ (3)
- $z_1 = (X_h + X_l)(Y_h + Y_l) - z_2 - z_0 = (101)(100) - 110 - 11 = 10100 - 1001 = 1011$ (11)
- Product = $z_2 \cdot 2^4 + z_1 \cdot 2^2 + z_0 = 1100000 + 101100 + 11 = 10001111 = 143$ ($11 \times 13 = 143$ ✓)
- b) 54×87 : $54 = 00110110$, $87 = 01010111$ (8-bit); split into 4-bit halves: $X_h = 0011$ (3), $X_l = 0110$ (6), $Y_h = 0101$ (5), $Y_l = 0111$ (7)
- $z_2 = 3 \times 5 = 15$; $z_0 = 6 \times 7 = 42$; $z_1 = (3 + 6)(5 + 7) - 15 - 42 = 9 \times 12 - 57 = 108 - 57 = 51$
- Product = $15 \times 256 + 51 \times 16 + 42 = 3840 + 816 + 42 = 4698$ ($54 \times 87 = 4698$ ✓)
- In binary: $4698 = 1001001011010$
- c) $X = 11011011$, $Y = 10111001$ setup: $X_h = 1101$, $X_l = 1011$, $Y_h = 1011$, $Y_l = 1001$
- Three recursive terms: $z_2 = X_h \cdot Y_h = 1101 \times 1011$; $z_0 = X_l \cdot Y_l = 1011 \times 1001$; $z_1 = (X_h + X_l)(Y_h + Y_l) - z_2 - z_0 = (11000)(10100) - z_2 - z_0$
- Final combination: Product = $z_2 \cdot 2^8 + z_1 \cdot 2^4 + z_0$